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Cite as: AIP Advances 12, 035019 (2022); <https://doi.org/10.1063/9.0000328>

Submitted: 01 November 2021 • Accepted: 05 December 2021 • Published Online: 14 March 2022

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Presented: 27 December 2021 • Submitted: 1 November 2021 •
Accepted: 5 December 2021 • Published Online: 14 March 2022



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Note: This paper was presented at the 15th Joint MMM-Intermag Conference.

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ABSTRACT

Electrical steels are still the materials of choice for large-scale transformers and most electric motors. Yet, they may present a nonhomogeneous magnetic nature which prevents describing accurately their anhysteretic magnetization with the Langevin-Weiss model. Although interpolation and extrapolation methods may be used to model any anhysteretic curve, a simple and physically-based model would be of great value for fundamental and applied research. Inspired in the law of partial volumes for gas mixtures, we proposed a law of partial magnetizations for magnetic mixtures. In a two-component system, the model leads to the double Langevin-Weiss function. We also introduced a graphical method and a fitting approach to analyze and model anhysteretic magnetization curves. A semi-log magnetization derivative plot is central to this end. We validated our strategy through well-motivated examples using published data on soft magnets. The single Langevin-Weiss function provided an accurate description of the magnetization of isotropic and anisotropic magnetically homogeneous materials: a soft ferrite and a nanocrystalline alloy, respectively. For modelling a magnetization transverse to the material's preferred direction, the key is to allow a negative molecular field constant. The double Langevin-Weiss function was suitable for less homogeneous materials, such as a grain-oriented electrical steel magnetized along the rolling direction and a non-oriented electrical steel. Moreover, a highly-grain-oriented electrical steel magnetized transverse to the rolling direction, which exhibits a constricted hysteresis loop, could be modeled in excellent agreement with data. The key for the latter, has been to allow an antiparallel arrangement of the mean magnetization of both components.

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INTRODUCTION

William Sturgeon constructed in England the first practical electromagnet using a soft iron core about two centuries ago.¹ Today, research on soft magnetic materials continues to be active and vibrant as they remain key components in the conversion, storage, and distribution of both energy and information.^{2,3} A simple and reliable description of the behavior of soft magnets is critical to the analysis and optimization of any robust electromagnetic device. Soft magnetic materials exhibit a nonlinear response of the isothermal magnetization M to an external magnetic field H , with a differential magnetic susceptibility $\chi = \partial M / \partial H$ that approaches to zero as $H \rightarrow \infty$. The magnetic state of soft magnets does not only depend on the applied field and temperature, but on the magnetic

history as well. However, computing the hysteretic behavior of magnetic components is still time consuming with current computers, so models usually only consider a single-valued, ideal, hysteresis-free magnetization curve, also called anhysteretic magnetization.^{4–7} Even when hysteresis needs to be simulated, the anhysteretic magnetization plays a crucial role, since it is the backbone of the popular Jiles-Atherton model^{8,9} implemented in modern software for lumped-element modeling¹⁰ and finite element analysis.¹¹

The anhysteretic curve may be calculated as the average curve of the left and right branches of a measured hysteresis loop,^{12,13} not only because of the technical challenges of its direct measurement,^{14–17} but because even its physical description remains elusive.^{18,19} Another challenge is that the curve can be rough due to noise introduced during measurements or digitization of scanned

data. Unphysical ripples in the derivative of the magnetization cause numerical instabilities that slow down any magnetic field solver and can even cause the lack of convergence.²⁰ In addition to the issue of the data noise, is the technical limitation that, at present, magnetic induction is measured only up to about 1.8 T.²¹ But in severe operating conditions, machines can show overfluxed regions that operate far beyond measured data, close to saturation.

Over the years, researchers have proposed several empiric formulae to model the anhysteretic magnetization, including rational, logarithmic, exponential, and inverse tangent functions, among others.^{22,23} But due to the lack of analytical functions that could describe –easily and accurately– the entire nonlinear magnetization process of soft magnetic materials, it has been claimed that, since the end of last century, these efforts have gone “out-of-fashion.”²¹ Instead, spline interpolation methods are usually used to achieve smooth magnetization data. These are combined with extrapolation methods to tackle the approach to saturation problem, facing the challenge of achieving reasonable results from a physical point of view.^{20,21}

Recently, we demonstrated that the Langevin-Weiss function is actually capable of describing the anhysteretic magnetization of any magnetically homogeneous soft magnet, over a broad range of applied fields.¹⁹ By allowing the molecular field constant (α) to be negative, we showed that the Langevin-Weiss function is suitable even for materials with uniaxial transverse anisotropy, previously considered an impossible scenario for the model.^{24,25} In the same study,¹⁹ we presented a deterministic strategy to find the model parameters that fit a $M-H$ data curve –rather than using traditionally stochastic methods within uncertain bounds. We followed such strategy to analyze different materials; an electrical steel, a soft ferrite, and a nanocrystalline alloy, noting that the least accurate fitting was obtained for the electrical steel. We suggested that this could be ascribed to its less homogeneous magnetic nature compared to the other materials. But electrical steels are still the materials of choice for large-scale transformers and for most electric motors, so a reliable assessment of their behavior would be of great value.

Here, we address the challenge of modeling, easily and accurately, the anhysteretic magnetization of less homogeneous materials, such as electrical steels. By recognizing that Weiss had inspired himself in van der Waals’ model of real gases for his theory of ferromagnetism (where elementary entities interact with each other), we asked whether the law of gas mixtures postulated by Amagat and Leduc could bring light to the problem of magnetic mixtures. With the aim of helping fundamental and applied investigations, we also looked for a method for readily revealing the magnetic homogeneity of a material and for an algorithm for optimizing the parameters of the model of magnetic mixtures.

FROM THE PARAMAGNETIC TO THE FERROMAGNETIC THEORY

In 1905, the French physicist Paul Langevin proposed a simple theory for paramagnetism,²⁶ treating matter as an assembly of N non-interacting entities per unit volume, each carrying a magnetic dipolar moment m_d . In the absence of an applied magnetic field, with the body in thermodynamic equilibrium, the moments are randomly oriented due to thermal agitation. Thus, the average

magnetic moment per unit volume –the magnetization M – is zero. In the presence of an external field H , an equilibrium establishes between thermal agitation and the tendency of the moments to align with H . Langevin defined the magnetization along the magnetic field direction as

$$M = Nm_d \langle \cos(\theta) \rangle \quad (1)$$

where the product Nm_d is the maximum possible value of M : the saturation magnetization M_s , which occurs when all moments are perfectly aligned with the external field. The contribution of each of the magnetic entities to the average magnetization is the projection of m_d on the direction of H , this is, the product of m_d and the thermal statistical average of $\cos(\theta)$, which follows the Boltzmann distribution. This reasoning leads to the so-called Langevin function $\mathcal{L}$ (the derivation may be found in Ref. 19):

$$M = M_s \left[\coth\left(\frac{H}{a}\right) - \left(\frac{a}{H}\right) \right] = M_s \mathcal{L}\left(\frac{H}{a}\right) \quad (2)$$

where the a parameter of the model depends on the Boltzmann constant k_B , on the absolute temperature T , on the vacuum magnetic permeability μ_0 , and on the magnetic moment of each entity m_d :

$$a = \frac{k_B T}{\mu_0 m_d} = \frac{N k_B T}{\mu_0 M_s} \quad (3)$$

The Langevin equation of state may also be written in terms of the reduced fields; the reduced magnetization, $m = M/M_s$, and the reduced magnetic field, $h = H/a$:

$$m = \mathcal{L}(h) \quad (4)$$

This theory is semiclassical or semiquantum in nature, because the magnetic moments m_d can point in any direction but have a fixed value, quantizing thus the system.^{20,21} If only discrete directions are allowed, the theory leads to the so-called Brillouin function. The equation compatible with quantum mechanics converges to the semiclassical Langevin function in the limit of the total angular momentum $J \rightarrow \infty$.²²

Meanwhile, in the Netherlands, Johannes D. van der Waals had proposed, in his thesis presented almost three decades earlier, a theory capable of explaining the behavior of fluids,²⁷ for which he would receive the Nobel Prize in 1910. In ideal gases, molecules do not attract or repel each other (electrostatic interactions are negligible). Therefore, when the system is at equilibrium, the pressure of the gas is equal to the external pressure. Van der Waals succeeded in explaining the behavior of real gases, quite simply and accurately, by making a couple of corrections to the equation of state of an ideal gas. One of these corrections, was to consider attractive dipolar interactions between molecules. Then, in equilibrium, the pressure of the gas would be equal to the external pressure plus an internal pressure, caused by those interactions.

Inspired in this idea, in 1907, Pierre E. Weiss explained ferromagnetism.²⁸ He stated that the magnetic field H acting on the magnetic entities, is the sum of the external field and an internal field, which he called the molecular field, caused by interactions between magnetic moments. Weiss proposed that this molecular field should be directly proportional to M (eventually expanding it

into higher order odd terms²⁹), without considering the spatial distribution of magnetic moments. Thus, with this simple equation of state:

$$M = M_S L\left(\frac{H + \alpha M}{a}\right) \quad (5)$$

that depends on the parameters M_S , a , and α , Weiss succeeded in describing the ideal magnetization or anhysteretic magnetization of ferromagnetic materials. What Weiss failed to explain, was the physical nature of the molecular field, because the magnetic dipole-dipole interactions were too weak to cause ferromagnetic alignment.³⁰

Almost 20 years later, in 1928, the German physicist Werner K. Heisenberg proposed a scenario of localized spins in which Coulomb interactions together with the Pauli exclusion principle, were sufficient to produce the same effects as the apparently large molecular field postulated by Weiss.³¹ This quantum phenomenon is known as exchange interaction and has no classical analog. Heisenberg explained that, under some considerations, the molecular field constant may be related to the exchange integral J_0 , by

$$\alpha = \frac{2z}{Ng^2\mu_B^2} J_0 \quad (6)$$

where all the other parameters are positive (the number of nearest neighbors z , the number of elementary magnetic entities per unit volume N , the Landé spectroscopic splitting factor g , and the Bohr magneton μ_B).³² J_0 , and thus α , would be positive for ferromagnetic alignment, and negative for antiparallel or antiferromagnetic alignment. But until the present day, nobody has been able to prove that J_0 must actually be positive in real ferromagnets, where orbital wave functions are more complex than in the scenario analyzed by Heisenberg.³³

In spite of the so-called sign problem of the exchange integral,³⁴ negative values of α have remained unexplored until recently in our previous work.¹⁹ In fact, we noticed that with negative molecular fields, the Langevin-Weiss model could describe the anhysteretic magnetization of ferromagnetic materials with strong induced transverse anisotropy. Before, researchers believed that the Langevin function was only suitable for isotropic materials.²⁴ Instead, they had proposed alternative functions to describe anisotropic materials that needed to be numerically integrated, increasing the computational cost.^{15,25} Yet, the counterintuitive negative molecular field –ferromagnetism is a cooperative phenomenon– would be explained by the fact that the argument of the Langevin function ($h = (H + \alpha M)/a$), is still being enhanced by the interplay between domain coupling, and magnetocrystalline and magnetoelectric anisotropies. In fact, a (which is in the denominator of h) is much smaller in soft magnets than in paramagnets, where any of these effects exist. Moreover, as already pointed out before, there is no theoretical demonstration that the exchange integral, and thus α , should actually be positive in real ferromagnets. By analyzing the effect of α on the shape of the anhysteretic magnetization,¹⁹ we found that the more negative α is, the wider the magnetic field range in which the magnetization is linear, this is, the wider the field range in which the magnetic susceptibility is uniform and the higher the anisotropy field is.

Last but not least, in the same work¹⁹ we addressed the challenge of optimizing the parameters of the Langevin-Weiss function

to fit data curves. As some of the bounds are a priori unknown and the problem presents local minima, traditional stochastic methods are time-consuming and may even fail to converge. To this end, we proposed a deterministic approach where the fitting variables are intrinsically bounded. By imposing that the modeled curve passes through two points of the data curve; the tip of the anhysteretic data curve, (H_{TIP}, M_{TIP}) , and some other point we called (H_X, M_X) , we were able to express M_S , a , and α , in terms of the reduced magnetization at the curve tip, m_{TIP} , and the magnetization at the other chosen constraint point, M_X :

$$\begin{cases} M_S = m_{TIP} M_{TIP} \\ \alpha = \frac{H_{TIP}}{M_{TIP}} \frac{\mathcal{L}^{-1}\left(\frac{M_X}{M_S}\right) - \frac{H_X}{H_{TIP}}}{\frac{M_X}{M_{TIP}} - \mathcal{L}^{-1}\left(\frac{M_X}{M_S}\right)} \\ a = \frac{H_{TIP} + \alpha M_{TIP}}{\mathcal{L}^{-1}(m_{TIP})} \end{cases} \quad (7)$$

The optimum m_{TIP} and M_X values can be found within intrinsic defined limits $-0 < m_{TIP} < 1$ and $0 < M_X < M_{TIP}$ – by minimizing the sum squared differences between the data and modeled curves. Optimizing only m_{TIP} and fixing M_X at, for example, about a 70% of M_{TIP} , allows to obtain a computationally cheaper and yet still good solution. The inverse Langevin function in Eq. (7) is evaluated using an approximant recently developed by Kröger,³⁵ due to its simultaneous simplicity and accuracy (maximum relative error $< 0.3\%$):

$$\mathcal{L}^{-1}(m) \cong \frac{3m - \frac{m}{5}(6m^2 + m^4 - 2m^6)}{1 - m^2} \quad (8)$$

EQUATION OF STATE FOR A MAGNETIC MIXTURE

The research group led by Jiles, has already reported an attempt in modeling two-phase magnetic materials³⁶ based on the Langevin-Weiss theory. In this study, they described all the model parameters (M_S , a , and α) as functions of H and of an exchange field, defined as the field at which a transition from one magnetic phase to the other would occur. Then, they used this anhysteretic magnetization as the backbone of a hysteretic magnetization model based on the Jiles-Atherton theory. The authors claimed that the parameters could be estimated with the help of a well-established algorithm. But the optimization process is not trivial due to the high number of fitting parameters, uncertain bounds and local minima and it was not described in the paper. We find the work to be highly relevant as they reported an excellent agreement of the magnetization of two different two-phase materials. However, these were reported in arbitrary units, without the fitting parameters, and without overlapping the simulated and measured curves. This suggests that the model is difficult to implement, while the physical basis and accuracy remain unclear.

Other studies have reported the technique of fitting magnetizations curves with the composition of two or more Langevin functions. In fact, this procedure is usually combined with microscopic studies for analyzing superparamagnets, either to determine multimodal grain size distributions^{37–40} or to identify components

in multiphase systems.⁴¹ Moreover, this approach has also been used by Steentjes et al. to describe the magnetization of soft magnetic steels, neglecting Weiss' molecular field, where "two passages of the magnetization process" are claimed to take place.⁴² Another strategy has been to describe the anhysteretic magnetization with a weighted sum of different anhysteretic functions, corresponding to different phases in a textured material: the Langevin-Weiss function to account for the isotropic phase and an alternative function proposed by Ramesh *et al.* to account for the anisotropic phase.⁴³

Noticing that Weiss had inspired himself in the theory of real gases proposed by van der Waals to account for the interactions between magnetic entities in ferromagnets, we looked into the theory of gas mixtures to bring light into the magnetic mixture scenario. In 1880 and 1898, Amagat^{44,45} and Leduc⁴⁶ found, from experiments on the mixture of gases, the law of additive component volumes: the volume of a mixture of gases is the sum of the volume of its components, each at the pressure and temperature of the mixture. The component volumes in a mixture are mathematical abstractions that have no physical interpretation. This model has shown to give satisfactory results even for mixtures of real gases up to pressures higher than with the law of additive component pressures,⁴⁷ i.e., Dalton's law for the mixture of gases published in 1802.⁴⁸

Inspired in this concept, we propose an equation of state of a mixture of magnetic entities, in which the magnetization of the mixture is the sum of the magnetization of its components, each at the magnetic field and temperature of the mixture:

$$M(H, T) = \sum_i M_i(H, T) = \sum_i M_{Si} m_i(H, T) \quad (9)$$

Thus, the saturation magnetization of the mixture is the sum of the saturation magnetization of its components. We propose that the reduced magnetization of each component follows the Langevin-Weiss function as

$$m_i = \frac{M_i}{M_{Si}} = L\left(\frac{H + \alpha_i M_i}{a_i}\right) = L\left(\frac{H + \alpha_i M_{Si} m_i}{a_i}\right) \quad (10)$$

Then, in a unit volume of a two-component mixture, the magnetization follows a double Langevin-Weiss function:

$$M = M_{S1} L\left(\frac{H + \alpha_1 M_{S1} m_1}{a_1}\right) + M_{S2} L\left(\frac{H + \alpha_2 M_{S2} m_2}{a_2}\right) \quad (11)$$

In this model, the unit volume is still treated as a uniform medium, dispensing the need of considering the spatial distribution of both sets of magnetic moments.

Computing the double Langevin-Weiss function to describe the magnetic response of a nonhomogeneous material may be an expensive operation for magnetic field solvers. Yet, the model provides a very attractive alternative to the combination of spline interpolation and blind extrapolation methods that cannot be overemphasized. By just storing the six model parameters, the physically-based equation of state (11) can generate a look-up-table – of the required precision – with a continuous and differentiable $M(H)$ curve, with its first derivative also continuous and differentiable that tends to zero as $H \rightarrow \infty$ (a look-up-table is an array that replaces runtime computation with a simpler array indexing operation).

DERIVATIVE AND SEMI-LOG DERIVATIVE PLOTS FOR ANHYSTERETIC MAGNETIZATION ANALYSIS

In this section, we present a method of graphical representation of the anhysteretic magnetization to serve as a tool for readily revealing the magnetic homogeneity of a material. In the following, we describe the set of plots shown in Fig. 1 that we propose for the anhysteretic magnetization analysis.

The first plot (Fig. 1(a)) is the typical linear representation of the $M - H$ curve. From this plot, we may observe that the Langevin-Weiss function, suitable for homogeneous materials, exhibits the following properties (already pointed out by Kokornaczyk and Gutowski²³): is single-valued, monotone, odd, equal to zero when H is zero, and saturates (i.e., $m \rightarrow 1$) as $H \rightarrow \infty$. Fig. 1(c) is another typical linear representation of the data, where the slope of the latter curve (i.e. $\partial M / \partial H$) is plotted vs H . In this plot, for the material to be described by the Langevin-Weiss function, the slope $\partial M / \partial H$ must be maximal at $H = 0$ and tend to zero as $H \rightarrow \infty$. In the case of constricted hysteresis loops, the average magnetization curve, however, has a local minimum at $H = 0$; the slope $\partial M / \partial H$ is maximal at $H > 0$. Constricted hysteresis loops have generally been associated with systems composed of a mixture of magnetic phases and have been observed, for example, in grain-oriented electrical steels under compressive stress^{36,49} and in highly grain-oriented electrical steels magnetized under arbitrary in-plane directions.⁵⁰

Fig. 1(b) is a representation of the $M - H$ curve; as Fig. 1(a), but with a logarithmic scale for H . Notice that there are no negative values for H on a semi-log plot, so only the first quadrant of the $M - H$ anhysteretic curve is analyzed (as opposed to Figs. 1(a) and 1(c), for which the entire $M - H$ plane can be plotted). The beauty of semilogarithmic plots (linear vertical axis, logarithmic horizontal axis) is that we can see details more easily for small fields as well as for the approach to saturation region. We may observe that the Langevin-Weiss function is "S"-shaped or sigmoidal in this plot. Steentjes et al. have used this semi-log plot to analyze the anhysteretic magnetization process of electrical steels.⁴² In fact, it was after observing a two-step sigmoid curve in this plot that they proposed fitting the data with the double Langevin function (as we previously stated, neglecting Weiss molecular field) and suggested the "two passages of the magnetization process."

The visual impression afforded by the semilogarithmic presentation can be enhanced by the introduction of the semi-log magnetization derivative (Fig. 1(d)), which represents the slope of the semilogarithmic plot (Fig. 1(c)) (actually, directly proportional to such slope, given that in Fig. 1(c) we used a common logarithmic scale for H and in Fig. 1(d) we plotted the derivative of M with respect to the natural logarithm of H – see Appendix A). For the Langevin-Weiss function plotted in the figure, we can readily identify a critical field (H_{cr}), at which the semi-log M derivative is maximum. From this consideration, we may find the following expressions for the parameter a of the Langevin-Weiss function (see Appendix A):

$$a = H_{cr} P \left(Q \pm \sqrt{Q^2 - 1} \right) \quad (12)$$

where,

$$P = \frac{\mathcal{L}'(h(H_{cr}))}{m(H_{cr}) - h\mathcal{L}'(h(H_{cr}))} \quad (13)$$

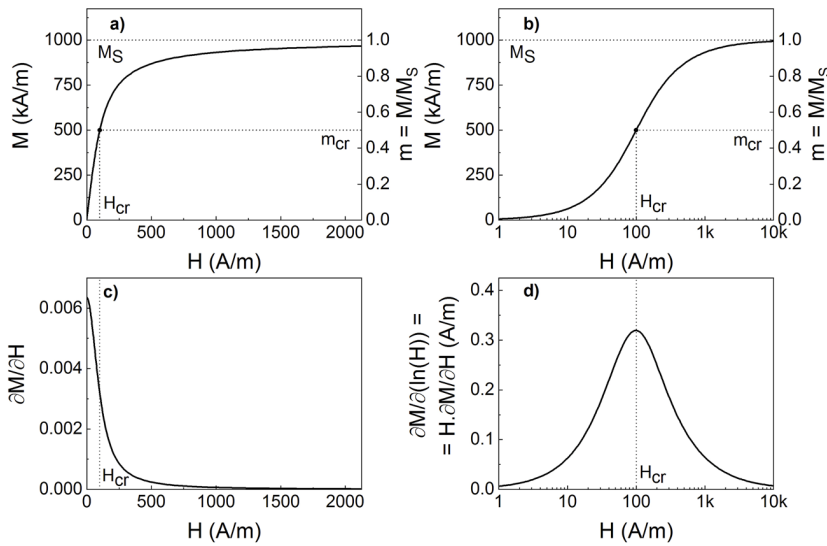


FIG. 1. Set of plots proposed for anhysteretic magnetization analysis: a) Typical linear M vs H plot, b) same as a) but with a logarithmic scale for H , c) Typical linear $\partial M/\partial H$ vs H plot, d) semi-log magnetization derivative vs H , with a logarithmic scale for H . The drawn curves correspond to the Langevin-Weiss function, suitable for homogeneous materials, with $M_S = 1000$ kA/m, $a = 71.66$ A/m, and $\alpha M_S = 57.38$ A/m. The semi-log magnetization derivative achieves its maximum at a critical field H_{cr} . In this example, $H_{cr} = 100$ A/m and the corresponding reduced magnetization is $m_{cr} = 0.5$.

$$Q = \left(\frac{m(H_{cr})}{\mathcal{L}'(h(H_{cr}))} \right)^2 \frac{(-\mathcal{L}''(h(H_{cr})))}{2(m(H_{cr}) - h(H_{cr})\mathcal{L}'(h(H_{cr})))} - 1 \quad (14)$$

In the above equations, $\mathcal{L}'(h)$ and $\mathcal{L}''(h)$ are the first and second derivatives, respectively, of the Langevin function with respect to h , while $h = \mathcal{L}^{-1}(m)$ (see Eqs. (4) and (8)). By knowing a , the product αM_S may be solved from the expression of the reduced magnetic field evaluated at H_{cr} :

$$\alpha M_S = \frac{h(H_{cr})a - H_{cr}}{m(H_{cr})} \quad (15)$$

The advantage of formulating a and the product αM_S in terms of H_{cr} and $m(H_{cr})$, is that it allows establishing an alternative approach to that presented in Ref. 19 for finding the parameters M_S , a , and α of the Langevin-Weiss function to fit a data curve. In this new approach, we also impose that the $M - H$ curve passes through (M_{TIP}, H_{TIP}) (i.e., the point at the tip of the data curve). The fitting variables are H_{cr} (for which a first approximation value together with reasonable bounds can be readily obtained from the semi-log M derivative plot), and $m(H_{cr})$, which must be positive and below unity. All the necessary steps of the algorithm are reported in Appendix B.

In this work, we minimized the mean orthogonal or diagonal distance error rather than the usual strategy of minimizing mean horizontal or vertical errors (see Appendix C). At the expense of a higher computational cost, we achieved more accurate results with this objective function. Also, since numerical differentiation typically results in noisy data, the objective function should compare the values of the $M - H$ curve, preferably in the $M - \log(H)$ representation (Fig. 1(b)), not its derivatives (Figs. 1(c) and (d)).

To determine the parameters of the double Langevin-Weiss function for a two-component system, we propose following an extended version of our new fitting approach reported in Appendix D. In the extended approach, we need to estimate the two critical fields from the semi-log M derivative plot, and impose

that the modeled curve passes through two points of the data curve: (M_{TIP}, H_{TIP}) and another point (M_X, H_X) , with H_X typically between H_{cr1} and H_{cr2} . The fitting variables are the two critical fields, H_{cr1} and H_{cr2} , H_X , and the reduced magnetization of each component at the corresponding critical field, $m_1(H_{cr1})$ and $m_2(H_{cr2})$.

In sum, the derivative and semi-log derivative plots (Figs. 1(c) and (d), respectively) serve as a diagnostic tool to make clear and more apparent the features of single- and multi-component magnetic systems. Then, the analyst can give a more conclusive interpretation of the anhysteretic magnetization data and readily recognize the critical fields that help finding the model parameters.

EXPERIMENTAL AND METHODS

In this work, we analyzed the anhysteretic behavior of different widely used soft magnetic materials: i) a grain-oriented electrical steel (M130-27s) measured parallel to the rolling direction, ii) a soft ferrite ($\text{Mn}_{0.51}\text{Zn}_{0.44}\text{Fe}_{2.05}\text{O}_4$) with very low anisotropy, iii) an iron-based nanocrystalline alloy ($\text{Fe}_{73.5}\text{Cu}_1\text{Nb}_3\text{Si}_{15.5}\text{B}_7$, under the trademark NANOPERM LM from Magnetec company) with in-plane uniaxial anisotropy induced transverse to the ribbon axis, measured along the ribbon axis, iv) a non-oriented electrical steel (JNEX-900), and v) a highly grain-oriented electrical steel (27-MOH) measured transverse to the rolling direction.

For the first three materials (grain-oriented electrical steel, soft ferrite, and nanocrystalline alloy), we used the quasistatic hysteresis loops distributed by Szweczyk through his Github repository.⁵¹ These measurements were carried out with a magnetizing field frequency of 0.5 Hz at room temperature, using the Epstein test frame for the steel, and toroidal cores for the ferrite and nanocrystalline samples (further measurement details were reported elsewhere^{15,52,53}). The data tables, originally provided as magnetic induction B vs the applied field H , were converted into M vs H through the relation $M = B/\mu_0 - H$. Then, the anhysteretic curve was

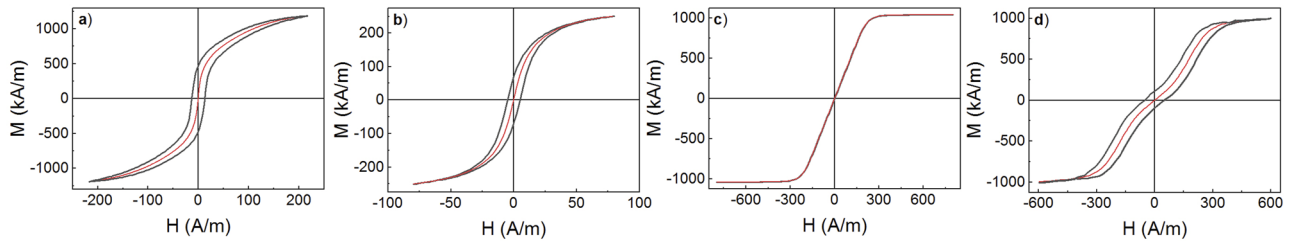


FIG. 2. Black curves show the quasistatic hysteresis loops of a) M130-27s grain-oriented electrical steel, measured parallel to the rolling direction,⁵¹ b) Mn-Zn ferrite,⁵¹ c) Fe-based nanocrystalline alloy with induced in-plane uniaxial anisotropy, measured transverse to the preferred direction,⁵¹ and d) highly grain-oriented electrical steel (27-MOH) measured transverse to the rolling direction.⁴² Red curves are the corresponding anhyseretic magnetizations calculated as the average curve of the left and right branches of the hysteresis loop.

calculated as the average curve of the left and right branches of the hysteresis loop. For a given value of M , we calculated the average value of H for both branches –as opposed to calculating for a given value of H , the average value of M for both branches– which is the method described by Bozorth²² to obtain an approximation of the “normal magnetization curve”, i.e. the first magnetization curve (see Fig. 2(a)–(c)). The anhyseretic curve of the non-oriented electrical steel was digitized from the graphs reported by Steentjes et al.⁴² using the software GetData Graph Digitizer.⁵⁴ For the highly grain-oriented electrical steel, we digitized the hysteresis loop reported by Baghel et al.⁵⁰ that was measured at 90° with respect to the rolling direction using a standard single sheet tester at 1 Hz. Then, the hysteresis loop reported as B vs H was converted into M vs H and averaged to obtain the anhyseretic curve as previously described (see Fig. 2(d)).

RESULTS AND DISCUSSION

We implemented the proposed graphical representation method (i.e., the set of plots reported in Fig. 1) to first analyze the magnetic homogeneity of the first three investigated materials, which have already been examined in our previous work:¹⁹ a grain-oriented electrical steel, measured parallel to the rolling direction, a soft ferrite, and an anisotropic nanocrystalline alloy, measured transverse to the preferred direction. Black dotted lines represent data curves and red solid lines, curves modeled with the single Langevin-Weiss function, using the set of parameters reported in each figure (Figs. 3-5). The bimodal characteristic of Fig. 3(d) reveals the magnetic inhomogeneity of the electrical steel, suggesting a two-component magnetic system. In fact, a much accurate description of the data was achieved when using a double Langevin-Weiss function (Fig. 6). These results support our presumption that the more

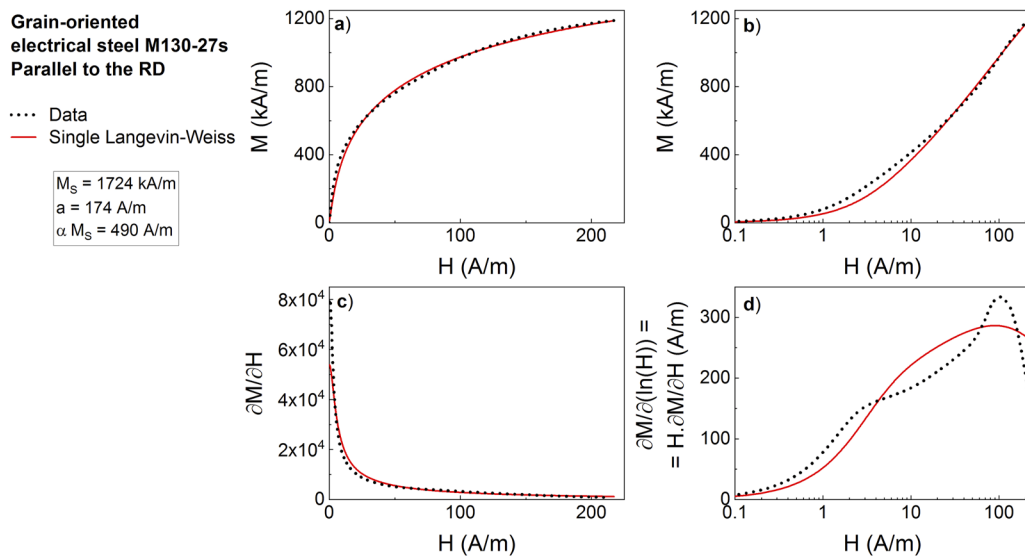


FIG. 3. Set of plots proposed for anhyseretic magnetization analysis of a grain-oriented electrical steel, measured parallel to the rolling direction: a) typical linear M vs H plot, b) same as a) but with a logarithmic scale for H , c) typical linear $\partial M/\partial H$ vs H plot, d) semi-log magnetization derivative vs H , with a logarithmic scale for H . Data was fitted using the single Langevin-Weiss function.

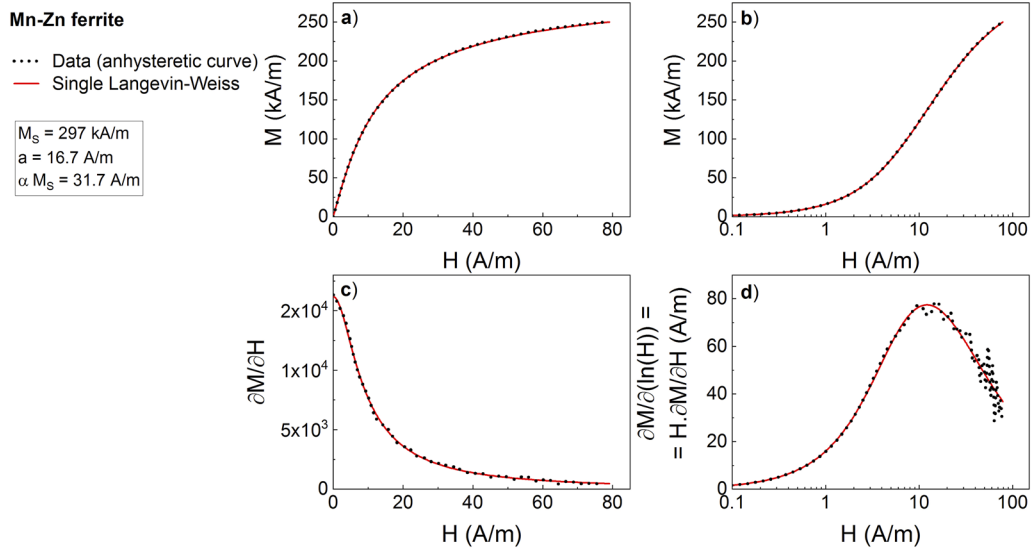


FIG. 4. Set of plots proposed for anhysteretic magnetization analysis of a soft ferrite: a) typical linear M vs H plot, b) same as a) but with a logarithmic scale for H , c) typical linear $\partial M / \partial H$ vs H plot, d) semi-log magnetization derivative vs H , with a logarithmic scale for H . Data was fitted using the single Langevin-Weiss function.

homogeneous nature of ferrite and nanocrystalline materials, as compared to that of steels, had led to a better accuracy of the single Langevin-Weiss model.¹⁹

We followed the same strategy to analyze and fit the anhysteretic curve of our fourth investigated material, a non-oriented electrical steel (Fig. 7). This curve had already been fitted by Steentjes et al.⁴² with a double Langevin function, capturing well the

two-component behavior. Still, adding the contribution of Weiss' molecular field further improved the fitting for this material. The improvement along the entire magnetization range can be better appreciated in Fig. 7(b).

Finally, we present the analysis on our last investigated material; a highly grain-oriented electrical steel, measured transverse to the rolling direction (Fig. 8). Fig. 8(d) exhibits only one halo, but the

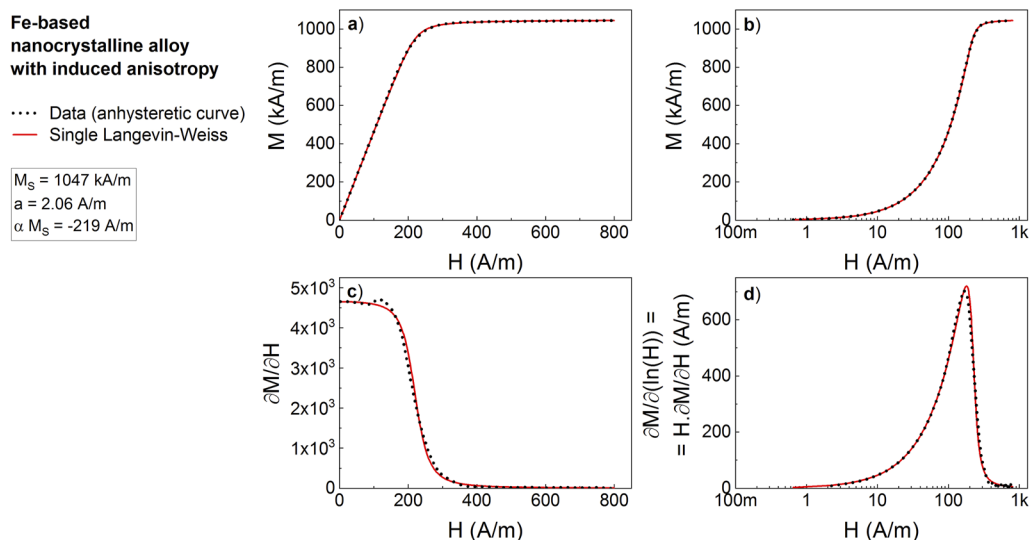


FIG. 5. Set of plots proposed for anhysteretic magnetization analysis of an anisotropic nanocrystalline alloy, measured transverse to the preferred direction: a) typical linear M vs H plot, b) same as a) but with a logarithmic scale for H , c) typical linear $\partial M / \partial H$ vs H plot, d) semi-log magnetization derivative vs H , with a logarithmic scale for H . Data was fitted using the single Langevin-Weiss function, allowing the molecular field constant to be negative.

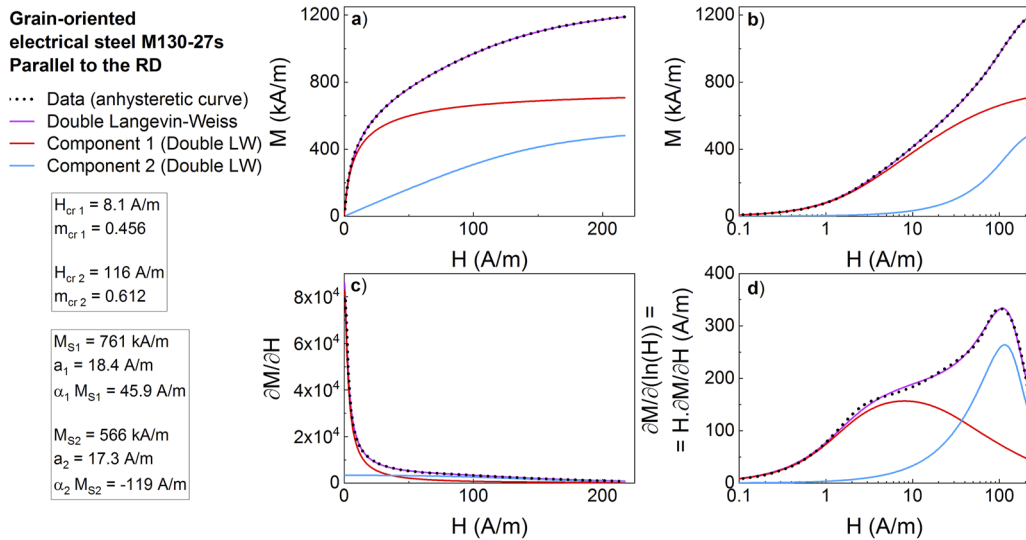


FIG. 6. Set of plots proposed for anhysteretic magnetization analysis of a grain-oriented electrical steel, measured parallel to the rolling direction: a) typical linear M vs H plot, b) same as a) but with a logarithmic scale for H , c) typical linear $\partial M/\partial H$ vs H plot, d) semi-log magnetization derivative vs H , with a logarithmic scale for H . Data was fitted using the double Langevin-Weiss function.

peak of Fig. 8(c) at $H > 0$ suggests a mixture of magnetic components. In fact, the anhysteretic curve could be accurately described by a double Langevin-Weiss function. The key has been to allow an antiparallel arrangement of both component magnetizations (notice that the saturation magnetization of one of the components is negative). For these cases, only one of the critical fields can be readily

identified from the semi-log derivative plot, but experience showed that a good seed for the other critical field in the optimization process is about $1/3$ or $1/2$ of the former. Further studies are needed to explore the relation between the model parameters and the direction of the applied field in these anisotropic magnetically nonhomogeneous materials.

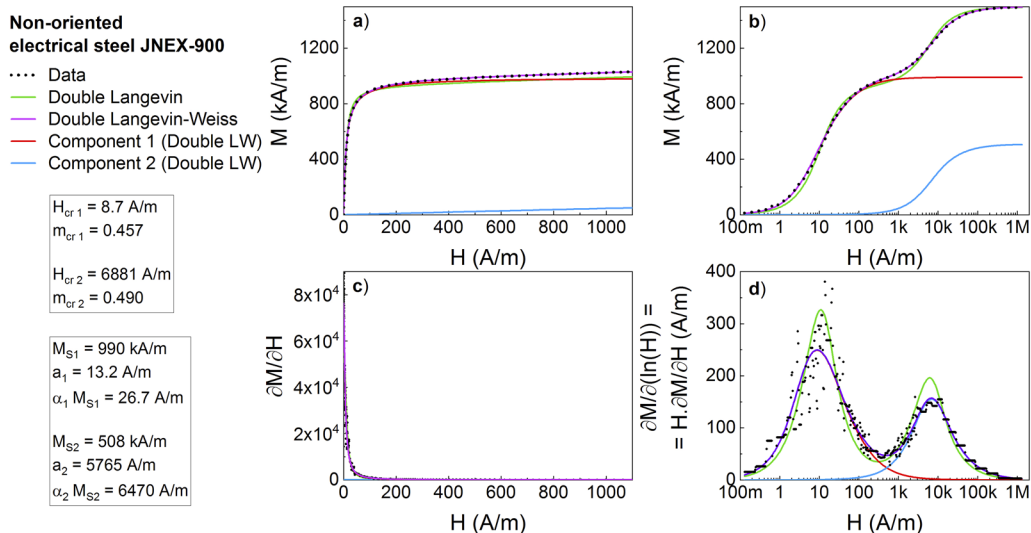


FIG. 7. Set of plots proposed for anhysteretic magnetization analysis of a non-oriented electrical steel. Data was fitted using the double Langevin-Weiss function: a) typical linear M vs H plot, b) same as a) but with a logarithmic scale for H , c) typical linear $\partial M/\partial H$ vs H plot, d) semi-log magnetization derivative vs H , with a logarithmic scale for H . We also plot the fitting using the double Langevin function reported by Steentjes et al.⁴²

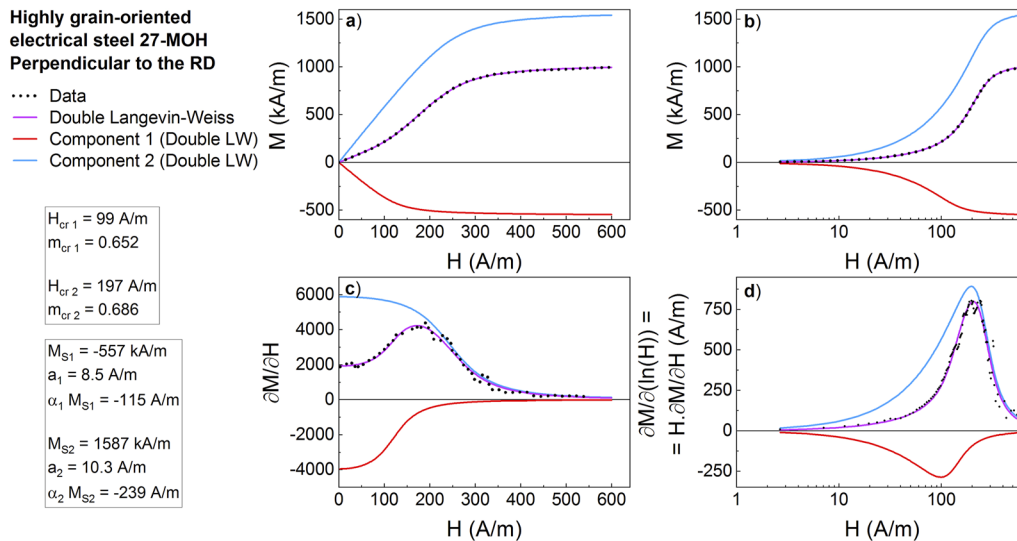


FIG. 8. Set of plots proposed for anhyseretic magnetization analysis of a highly-grain-oriented electrical steel, measured transverse to the rolling direction: a) typical linear M vs H plot, b) same as a) but with a logarithmic scale for H , c) typical linear $\partial M/\partial H$ vs H plot, d) semi-log magnetization derivative vs H , with a logarithmic scale for H . Data was fitted using the double Langevin-Weiss function, allowing antiparallel magnetic components.

CONCLUSIONS

In this work, we extended the Langevin-Weiss theory, which describes the anhyseretic magnetization of homogeneous soft magnets –isotropic or anisotropic–, to assess the behavior of magnetically non-homogeneous materials. Inspired by the law of mixture of gases postulated by Amagat and Leduc, we proposed a law of mixture of magnetic moments that, for a two-component system, leads to a double Langevin-Weiss function. We also introduced a method of graphical representation, including derivative and semi-log derivative plots, for analyzing the anhyseretic magnetization and readily revealing the magnetic homogeneity of a material.

By identifying the critical field of the semi-log M derivative plot, we developed an alternative approach from that presented in Ref. 19 for finding the Langevin-Weiss model parameters. The former approach was simpler and faster to compute, while the new one could be extended for finding the parameters of the double Langevin-Weiss function. The typical fitting approach has been to directly search for the optimum model parameters. This is a challenging task because the optimization problem presents local minima and several uncertain bounds. Conversely, our strategy focuses in expressing the model parameters in terms of variables that are optimized within known limits.

We validated our approach through some well-motivated examples using published data on soft magnets. The results either improve previous ones or are completely new. In fact, the theory provides a very good description, not only of anhyseretic magnetizations exhibiting a bimodal semi-log M derivative plot, but even of anhyseretic magnetizations exhibiting a peak at $H > 0$ in the linear M derivative plot, reconciling seemingly disparate scenarios. The key for the latter is to allow an antiparallel or antiferromagnetic

arrangement of both sets of magnetic components. Further studies are needed to better understand the physics behind this theory.

To speed up runtime computation of magnetic field solvers, the double Langevin-Weiss function may be used to generate a look-up table of the anhyseretic curve of magnetically nonhomogeneous materials. The model guarantees the lack of unphysical ripples in the derivative of the magnetization and a physically reasonable approach to saturation.

This extension on the Langevin-Weiss theory opens exciting opportunities for further understanding the magnetization process of soft magnets, as well as for improving the design and performance of electromagnetic devices.

ACKNOWLEDGMENTS

This work was supported by CONICET, Universidad de Buenos Aires (Grant number UBACyT 20020190200001BA), and ANPCyT (Grant number PICT-2019-2019-02122).

AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

All authors contributed equally to this work.

DATA AVAILABILITY

The data that support the findings of this study are available within the article.

APPENDIX A: DERIVATION OF EQUATIONS FOR THE ANALYSIS OF THE LANGEVIN-WEISS FUNCTION

In this section, we derive Eqs. (12–15). The idea behind this approach, is that the semi-log magnetization derivative exhibits a maximum at a critical field we call H_{cr} . This sets a relation between the Langevin-Weiss model parameters that, together with the definition of reduced magnetization, may be solved for a and for the product αM_S . To derive these equations, we first need to define some functions and their derivatives.

The reduced magnetization ($m = M/M_S$) and the reduced magnetic field ($h = (H + \alpha M)/a$) are related by the Langevin function as

$$m = L(h) \quad (A1)$$

which is

$$L(h) = \coth(h) - 1/h \quad (A2)$$

In the vicinity of $h = 0$, the Langevin function suffers from loss of significance, and it is numerically better to approximate it by a truncation of its Taylor series expansion about the origin:

$$L(h) = \frac{h}{3} - \frac{1}{45}h^3 + \frac{2}{945}h^5 - \dots \quad (A3)$$

The first and second derivatives of the Langevin function with respect to h are

$$\mathcal{L}'(h) = \frac{\partial L(h)}{\partial h} = -\csc^2(h) + 1/h^2 \quad (A4)$$

$$\mathcal{L}''(h) = \frac{\partial \mathcal{L}'(h)}{\partial h} = 2 \coth(h) \csc^2(h) - 2/h^3 \quad (A5)$$

being their Taylor series expansions about the origin:

$$L'(h) = \frac{1}{3} - \frac{h^2}{15} + \frac{2}{189}h^4 - \dots \quad (A6)$$

$$\mathcal{L}''(h) = -\frac{2}{15}h + \frac{8}{189}h^3 - \frac{2}{225}h^5 + \dots \quad (A7)$$

The magnetic susceptibility, this is, the first derivative of M with respect to H is

$$\frac{\partial M}{\partial H} = \frac{\partial(M_S m)}{\partial H} = M_S \frac{\partial m}{\partial H} = M_S \frac{\partial m}{\partial h} \frac{\partial h}{\partial H}$$

$$\frac{\partial M}{\partial H} = M_S \mathcal{L}'(h) \frac{1}{a} \left(1 + \alpha \frac{\partial M}{\partial H} \right)$$

$$\frac{\partial M}{\partial H} \left(1 - \frac{\alpha M_S \mathcal{L}'(h)}{a} \right) = \frac{M_S \mathcal{L}'(h)}{a}$$

$$\frac{\partial M}{\partial H} = \frac{M_S \mathcal{L}'(h)/a}{1 - \alpha M_S \mathcal{L}'(h)/a} \quad (A8)$$

The first derivative of h with respect to H is

$$\begin{aligned} \frac{\partial h}{\partial H} &= \frac{1}{a} \left(1 + \alpha \frac{\partial M}{\partial H} \right) = \frac{1}{a} \left(1 + \alpha \left(\frac{M_S \mathcal{L}'(h)/a}{1 - \alpha M_S \mathcal{L}'(h)/a} \right) \right) \\ &= \frac{1}{a} \left(\frac{1 - \alpha M_S \mathcal{L}'(h)/a + \alpha M_S \mathcal{L}'(h)/a}{1 - \alpha M_S \mathcal{L}'(h)/a} \right) \end{aligned}$$

$$\frac{\partial h}{\partial H} = \frac{1}{a - \alpha M_S \mathcal{L}'(h)} \quad (A9)$$

We define the semi-log M derivative as the slope of the data when plotted on semi-log coordinates, which is the derivative of M with respect to the logarithm of H . This is, $\partial M / \partial(\log H)$ using the common logarithm, and $\partial M / \partial(\ln H)$ using the natural logarithm, both related by

$$\frac{\partial M}{\partial(\log H)} = \ln(10) \frac{\partial M}{\partial(\ln H)} \cong 2.303 \frac{\partial M}{\partial(\ln H)} \quad (A10)$$

$\partial M / \partial(\ln H)$ can be calculated as follows:

$$\frac{\partial M}{\partial(\ln H)} = \left(\frac{\partial(\ln H)}{\partial M} \right)^{-1} = \left(\frac{1}{H} \frac{\partial H}{\partial M} \right)^{-1} = H \frac{\partial M}{\partial H} \quad (A11)$$

$$\frac{\partial M}{\partial(\ln H)} = H \frac{M_S \mathcal{L}'(h)/a}{1 - \alpha M_S \mathcal{L}'(h)/a} \quad (A12)$$

Both $\partial M / \partial(\log H)$ and $\partial M / \partial(\ln H)$ are maximal at the same critical field, denoted herein as H_{cr} . This means that their derivative with respect to H must be zero at H_{cr} . We first seek for the expression of $\frac{\partial}{\partial H} \left(\frac{\partial M}{\partial(\ln H)} \right)$:

$$\frac{\partial}{\partial H} \left(\frac{\partial M}{\partial(\ln H)} \right) = \frac{M_S \mathcal{L}'(h)/a}{1 - \alpha M_S \mathcal{L}'(h)/a} + H \frac{\partial}{\partial h} \left(\frac{M_S \mathcal{L}'(h)/a}{1 - \alpha M_S \mathcal{L}'(h)/a} \right) \frac{\partial h}{\partial H}$$

$$\begin{aligned} \frac{\partial}{\partial H} \left(\frac{\partial M}{\partial(\ln H)} \right) &= \frac{M_S \mathcal{L}'(h)/a}{1 - \alpha M_S \mathcal{L}'(h)/a} + H \left(\frac{a M_S \mathcal{L}''(h)}{(a - \alpha M_S \mathcal{L}'(h))^2} \right) \\ &\quad \times \left(\frac{1}{a - \alpha M_S \mathcal{L}'(h)} \right) \end{aligned}$$

$$\frac{\partial}{\partial H} \left(\frac{\partial M}{\partial(\ln H)} \right) = \frac{M_S \mathcal{L}'(h)}{a - \alpha M_S \mathcal{L}'(h)} + \frac{H a M_S \mathcal{L}''(h)}{(a - \alpha M_S \mathcal{L}'(h))^3} \quad (A13)$$

Then, by setting it equal to zero at $H = H_{cr}$ (which can be readily identified from the plot of the semi-log M derivative –Fig. 1.d), we obtain a relation between the Langevin-Weiss model parameters:

$$\frac{\partial}{\partial H} \left(\frac{\partial M}{\partial(\ln H)} \right) \Big|_{H=H_{cr}} = 0$$

$$\alpha M_S = \frac{a - \sqrt{-H_{cr} a \mathcal{L}''(h(H_{cr})) / \mathcal{L}'(h(H_{cr}))}}{\mathcal{L}'(h(H_{cr}))} \quad (A14)$$

Evaluating the reduced magnetization at $H = H_{cr}$, we can obtain another relation between the Langevin-Weiss model parameters:

$$h|_{H=H_{cr}} = \frac{H + \alpha M_S m}{a} \Big|_{H=H_{cr}}$$

$$\alpha M_S = \frac{h(H_{cr})a - H_{cr}}{m(H_{cr})} \quad (\text{A15})$$

We can then equate the two latter equations, (A14) and (A15), and solve for a . After some algebra, two possible solutions for a may be found:

$$a = H_{cr} P \left(Q \pm \sqrt{Q^2 - 1} \right) \quad (\text{A16})$$

where,

$$P = \frac{\mathcal{L}'(h(H_{cr}))}{m(H_{cr}) - h\mathcal{L}'(h(H_{cr}))} \quad (\text{A17})$$

$$Q = \left(\frac{m(H_{cr})}{\mathcal{L}'(h(H_{cr}))} \right)^2 \frac{(-\mathcal{L}''(h(H_{cr})))}{2(m(H_{cr}) - h(H_{cr})\mathcal{L}'(h(H_{cr})))} - 1 \quad (\text{A18})$$

APPENDIX B: ALGORITHM FOR FINDING THE PARAMETERS OF THE LANGEVIN-WEISS FUNCTION

In this section, we describe an alternative approach to that presented in Ref. 19 for finding the parameters M_S , a , and α of the Langevin-Weiss function. As in Ref. 19, the approach optimizes variables within certain bounds, as opposed to directly fitting M_S , a , and α , where we do not know the lower bound of α and the upper bounds of all the variables. This approach is based on identifying the critical field H_{cr} in the semi-log M derivative plot introduced in the present work and on imposing that the $M - H$ curve passes through (M_{TIP}, H_{TIP}) (i.e., the point at the tip of the data curve).

Step 1: Identify the critical field, H_{cr} , from the semi-log M derivative plot (see Fig. 1; in step 8, H_{cr} may be optimized).

Step 2: Propose the reduced magnetization at H_{cr} , $m(H_{cr})$, which must be positive and below unity. Experience has shown that 0.5 is typically a good starting seed (in Step 8, the optimum $m(H_{cr})$ will be found).

Step 3: Calculate $h(H_{cr}) = \mathcal{L}^{-1}(m(H_{cr}))$ using Kröger's approximant of the inverse Langevin function:

$$\mathcal{L}^{-1}(m) \cong \frac{3m - \frac{m}{5}(6m^2 + m^4 - 2m^6)}{1 - m^2} \quad (\text{B1})$$

Step 4: Compute P and Q (Eqs. (A17) and (A18), respectively) to calculate a (Eq. (A16)) and then the product αM_S (Eq. (A15)).

Step 5: Calculate the reduced magnetization at the tip of the $M - H$ data curve to fit, $m(H_{TIP})$, by finding the roots of the function below that follows from the Langevin function (A1):

$$f(m) = L\left(\frac{H + \alpha M_S m}{a}\right) - m = 0 \quad (\text{B2})$$

Notice that $H = H_{TIP}$, and that a and αM_S have been calculated in Step 4. The bisection method can be used for this

purpose, as it is a very simple and robust root-finding method that applies to any continuous functions for which one knows two values with opposite signs; in this case being $0 < m(H_{TIP}) < 1$. As it is also relatively slow, it may be combined in a hybrid algorithm with the Newton-Raphson method for a more rapid convergence (see the routine RTSafe in Ref. 55).

Step 6: Identify the tip magnetization, M_{TIP} , and calculate M_S as

$$M_S = M_{TIP}/m(H_{TIP}) \quad (\text{B3})$$

Step 7: If wanted, calculate α as

$$\alpha = \alpha M_S / M_S \quad (\text{B4})$$

but notice this is not really necessary for calculating the anisotropic magnetization M at a given applied field H , or vice versa.

Step 8: Explore the search domain ($0 < m(H_{cr}) < 1$; reasonable limits of H_{cr} should be chosen from the semi-log M derivative plot) and locate the best solution, which minimizes the objective function (see Appendix C). A possible nonlinear solving method for this purpose is the generalized reduced gradient nonlinear solver. Notice that (Eq. (A16)), used in Step 4 to calculate a , has two solutions. The search domain may be thus explored twice, once for the smallest a solution, and once for the largest. The smallest a describes the narrowest halo in the semilogarithmic plot of $\frac{\partial M}{\partial(\ln H)}$ vs H .

APPENDIX C: OBJECTIVE FUNCTION

Fitting data in the linear plot (M vs H , Fig. 1.a) may diminish phenomena of interest at weak applied fields for materials with high initial magnetic susceptibility. Experience has shown that it is convenient to match instead the semi-log plot (M vs $\ln(H)$, Fig. 1.b).

By calling Y the vertical axis, X the vertical axis, and N the point at the tip of the magnetization curve, rather than minimizing the normalized root-mean squared error in X at constant Y :

$$NRMSE_X = \sqrt{\sum_{i=0}^N (X_i^{data}(Y_i^{data}) - X^{sim}(Y_i^{data}))^2 / X_N^{data}(Y_N^{data})} \quad (\text{C1})$$

or the normalized root-mean squared error in Y at constant X :

$$NRMSE_Y = \sqrt{\sum_{i=0}^N (Y_i^{data}(X_i^{data}) - Y^{sim}(X_i^{data}))^2 / Y_N^{data}(X_N^{data})} \quad (\text{C2})$$

at the expense of a higher computational cost, we obtained a better fit by minimizing the mean orthogonal distance error (also known as diagonal distance error) in a normalized plane $x - y$ ($x = X/X_N$ and $y = Y/Y_N$):

$$MOE = \sqrt{\sum_{i=0}^N \Delta o_i^2} \quad (\text{C3})$$

where Δo_i follows from Fig. 9:

$$\Delta o_i = \Delta y_i \cos(\alpha) = \Delta y_i \cos(\text{atan}(\Delta y_i / \Delta x_i)) \quad (\text{C4})$$

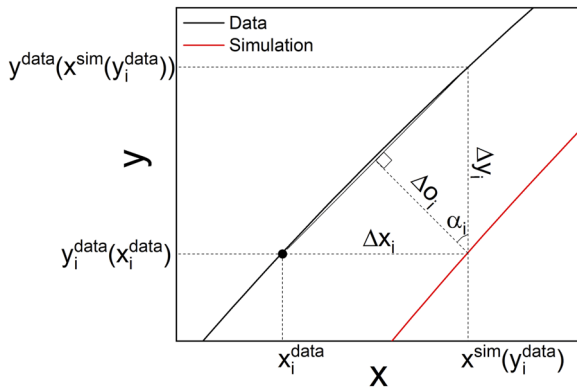


FIG. 9. Data and simulated curves pointing out the parameters used in the objective function that minimizes the mean orthogonal or diagonal distance error.

with:

$$\begin{cases} \Delta x_i = \left| X^{\text{sim}}(Y_i^{\text{data}}) - X_i^{\text{data}} \right| / X_N^{\text{data}} \\ \Delta y_i = \left| Y^{\text{data}}(X^{\text{sim}}(Y_i^{\text{data}})) - Y_i^{\text{data}}(X_i^{\text{data}}) \right| / Y_N^{\text{data}} \end{cases} \quad (\text{C5})$$

$X^{\text{sim}}(Y_i^{\text{data}})$ and $Y^{\text{data}}(X^{\text{sim}})$ may be estimated by interpolation of the $X^{\text{sim}} - Y^{\text{sim}}$ and $X^{\text{data}} - Y^{\text{data}}$ curves, respectively.

APPENDIX D: ALGORITHM FOR FINDING THE PARAMETERS OF THE DOUBLE LANGEVIN-WEISS FUNCTION

In this section, we describe an approach for systematically finding the parameters M_{Si} , a_i , and α_i of the two components of the double Langevin-Weiss function. The approach optimizes variables within certain bounds, as opposed to directly fitting M_{Si} , a_i , and α_i , where all but the lower bound of a_i are uncertain. This approach is based on analyzing the critical fields $H_{cr,i}$ in the semi-log M derivative introduced in the present work, and on imposing that the $M - H$ curve passes through two points of the data curve: (M_{TIP}, H_{TIP}) and another point we call (M_X, H_X) , where H_X is typically between H_{cr1} and H_{cr2} .

Steps 1-5: Follow Steps 1 through 5 from Appendix B to identify $H_{cr,1}$ and $H_{cr,2}$, propose $m_1(H_{cr,1})$ and $m_2(H_{cr,2})$, and calculate a_i together with the product $\alpha_i M_{Si}$ for each magnetic component.

Step 6: Identify the tip of the $M - H$ data curve, (M_{TIP}, H_{TIP}) , and choose another constraint point, (M_X, H_X) . Typically, H_X should be between $H_{cr,1}$ and $H_{cr,2}$ (in step 8, H_X may be optimized). The corresponding magnetization, M_X , can be estimated by interpolation of the data curve. Next, calculate the reduced magnetizations of both components at both fields $-m_1(H_X)$, $m_2(H_X)$, $m_1(H_{TIP})$, and $m_2(H_{TIP})$ —by numerically finding the roots of Eq. (B2) (B.2). Then, M_{S1} and M_{S2} can be solved from the following system of equations:

$$\begin{cases} M_{TIP} = m_1(H_{TIP})M_{S1} + m_2(H_{TIP})M_{S2} \\ M_X = m_1(H_X)M_{S1} + m_2(H_X)M_{S2} \end{cases} \quad (\text{D1})$$

as,

$$\begin{cases} M_{S1} = \frac{m_2(H_X)M_{TIP} - m_2(H_{TIP})M_X}{m_2(H_X)m_1(H_{TIP}) - m_2(H_{TIP})m_1(H_X)} \\ M_{S2} = \frac{m_1(H_X)M_{TIP} - m_1(H_{TIP})M_X}{m_1(H_X)m_2(H_{TIP}) - m_1(H_{TIP})m_2(H_X)} \end{cases} \quad (\text{D2})$$

Step 7: If wanted, calculate α_i for both components as in Step 7 of Appendix B.

Step 8: Explore the search domain ($0 < m_1(H_{cr,1}) < 1$; $0 < m_2(H_{cr,2}) < 1$; reasonable limits of $H_{cr,i}$ should be chosen from the semi-log derivative plot of M ; $0 < H_X < H_{TIP}$ but, typically, $H_{cr,1} < H_X < H_{cr,2}$). Use a nonlinear solving method (e.g. the generalized reduced gradient nonlinear solver) to locate the best solution, which minimizes the objective function (Appendix C). Notice that Eq. (A16), used in Step 4 to calculate a has two solutions. The search domain may be thus explored $2^2 = 4$ times; once for each combination of a_1 and a_2 . Noticing that the smallest value of a_i describes the narrowest halo in the semilogarithmic plot of $\frac{\partial M_i}{\partial(\ln H)}$ vs H , with some experience, the number of times needed to explore the search domain might be narrowed down to only one by looking at the semilogarithmic data plot of $\frac{\partial M}{\partial(\ln H)}$ vs H .

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